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SARDAR PATEL UNIVERSITY

M.Sc. CA&IT Examination Nov - 2019

1st Semester NC (CBCS) (Regular)

Monday , Date: 25-11-2019

Session : Morning Time : 10:00 A.M. TO 12: 00 P.M.

Course Code: PS01EIIT31

Course Title : Image Processing **Total Marks: 70**

Q.1 Multiple Choice Questions. (Attempt all) [08]

1. The Process of extracting information from the digital image is known as _____.
   - A. Image Compression
   - B. Image Segmentation
   - C. Image Analysis
   - D. Image Restoration

2. Which of the following are photoreceptor cells present in retinal surface of the eye?
   - A. Rods and cones
   - B. Rods and fovea
   - C. Iris and Pupil
   - D. Choroid and Cortex

3. Statistical information about pixel brightness is represented by _______.
   - A. Zooming
   - B. Pseudo coloring
   - C. Histogram
   - D. Edge detection

4. 2D Fourier transform and its inverse are indefinitely ________.
   - A. Periodic
   - B. Non-linear
   - C. Linear
   - D. symmetric

5. Noise that corresponds to extreme values of impulse, with intensity 0 for dark and 255 for bright is known _________ noise.
   - A. Gamma
   - B. Random
   - C. Zero and Negative
   - D. Salt and Pepper

6. The purpose of image restoration is to ________________.
   - A. Blur original image
   - B. Add noise to original image
   - C. Recover degraded image
   - D. Slice original image

7. Consider a string ***aaaabbbbbb*** is reduced implicitly to ________.
   - A. {(a,4),(b,6)}
   - B. {(a),(b),(4),(6)}
   - C. {(b,6),(a,4)}
   - D. {(4),(6),(a),(b)}

8. Savings percentage is defined as ________________.
   - A. $1 - \frac{\text{Message size after compression}}{\text{Code size before compression}}$
   - B. $\frac{\text{Message size after compression}}{\text{Code size before compression}}$
   - C. $1 - \frac{\text{Code size before compression}}{\text{Message size after compression}}$
   - D. $\frac{\text{Code size before compression}}{\text{Message size after compression}}$

Q.2 Short Questions. (Attempt any seven) [14]

1. Define : (i) Digital Image (ii) Pixel
2. List application areas of Digital Image Processing.
3. What is Image Enhancement ? List Image Enhancement Techniques.
4. What is Image Smoothing?
5. List properties of two dimensional Fourier transformation.
6. List different filtering methods used for Image Restoration process.
7. List different types of Redundancy.
8. What is Data Compression ? List types of data encountered in Image Processing.
9. State reasons essential for Data Compression. Explain any one in brief.

(P.T.O)

Q.3[a] Draw and Explain the structure of Human Eye and the Visual Preliminaries in detail. [06]

Q.3[b] Explain the components of Digital Image processing in detail. [06]

OR

Q.3[b] Explain Fundamental steps in Digital Image Processing. [06]

Q.4[a] Explain Two Dimensional Fourier transformation in detail. [06]

Q.4[b] Explain concept of Image Sharpening in brief. [06]

Consider the following figure where each small rectangle represents a pixel and the value inside it is gray level at that pixel. Hence whole array represents digital image ***g(r,c)*** of size 5 x 5 . The center pixel g(2,2) is marked by underline. Sharpen the image in spatial domain using Laplacian 4-connectivity approach, calculate value of ***g'(2,2).***

0	1	0	6	7
2	0	1	6	5
1	1	$\underline{7}$	5	6
1	0	6	6	5
2	5	6	7	6

OR

Q.4[b] What is Contrast Intensification ? [06]

Given ***m*** is the gray level of input 8-level image which has to be transformed to ***l*** by linear stretching. Frequency table for input gray level and $\mathbf{n_i}$ – number of pixel having $\mathbf{i^{th}}$ input gray level are given as below :

i	0	1	2	3	4	5	6	7
$\mathbf{n_i}$	0	0	a	b	c	d	e	0

Find the frequency table for output graylevels.

Q.5[a] What is Image Restoration? Explain model for Image Degradation/Restoration Process. [06]

Q.5[b] Explain Translation and Scaling geometric transformations in three-dimensional cartesian coordinate system. [06]

OR

Q.5[b] Explain any two noise models used for restoring image. [06]

Q.6[a] Differentiate between Lossy and Lossless compression algorithms. [06]

Q.6[b] List classification of Image Compression algorithms and explain any two in detail. [06]

OR

Q.6[b] Explain Image Compression scheme and Model with labeled diagrams. [06]

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